

Adiposity as a chronic relapsing disease: pathophysiology, comorbidity burden, and the evolving therapeutic landscape

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Abstract Obesity is increasingly characterised as a chronic, relapsing disease of energy regulation rather than a behavioural failing. This review synthesises current evidence across four domains. In pathophysiology, attenuated afferent leptin and insulin signalling raises the defended body-weight set point, and the counter-regulatory response to energy restriction provides a mechanistic account of the regain trajectories observed across behavioural programmes. In epidemiology, age-standardised prevalence has approximately doubled since 1990, with the steepest projected increases now in low- and middle-income settings. In comorbidity, the association with incident type 2 diabetes remains the strongest and most consistently replicated, while musculoskeletal and psychiatric burden are frequently underweighted in conventional risk calculators. In management, incretin-pathway pharmacotherapy has achieved effect sizes approaching those of bariatric surgery within a single product generation, though real-world persistence falls substantially below trial observation. We argue that the binding constraint is no longer the availability of effective treatment but the capacity of health systems to deliver it for the duration a chronic disease requires.

Keywords: obesity; adiposity; energy homeostasis; incretin; comorbidity; health economics

1 Introduction

Obesity is now understood less as a failure of individual restraint than as a chronic, relapsing, progressive disease of energy regulation.^{1–4} The reframing is not merely semantic: it changes what counts as treatment success, how long therapy is expected to continue, and which outcomes justify reimbursement.^{5,6} Regulatory and professional bodies across several jurisdictions have adopted disease framing over the past decade, though implementation in routine care remains uneven.^{7–9, 12}

Three developments motivate the present review. First, prevalence has continued to rise in almost every reporting region despite three decades of public health intervention.^{10, 11, 13–16} Second, mechanistic work on the hypothalamic control of energy balance has produced a coherent account of why weight loss achieved through energy restriction is so reliably followed by regain.^{17–21} Third, pharmacotherapy has moved, within a single product generation, from mean placebo-subtracted reductions of three to five per cent to reductions exceeding fifteen per cent.^{22–26} The third development in particular has outpaced the care pathways intended to deliver it.

This review synthesises evidence across pathophysiology, comorbidity burden, assessment, and management, with attention to the gap between trial efficacy and realised benefit in routine practice.^{27–30}

2 Definitions and Assessment

Body mass index remains the dominant screening metric by virtue of availability rather than accuracy.^{31–33} It does

Table 1: Conventional body mass index classification and associated screening considerations. Thresholds are those in widest use; several groups recommend ancestry-specific adjustment.^{34,36,39}

Category	BMI (kg/m ²)	Adjunct measure
Underweight	<18.5	Body composition
Normal range	18.5–24.9	—
Overweight	25.0–29.9	Waist circumference
Obesity, class I	30.0–34.9	Waist-to-height ratio
Obesity, class II	35.0–39.9	Metabolic staging
Obesity, class III	≥40.0	Metabolic staging

not distinguish adipose from lean mass, performs inconsistently across ancestry groups, and misclassifies at the extremes of stature.^{31, 34–37} Waist circumference and waist-to-height ratio improve discrimination of visceral adiposity at negligible additional cost, and several groups now recommend their routine use alongside BMI rather than in place of it.^{38–41}

Functional staging systems, which grade severity by the presence of adiposity-related complications rather than by anthropometry alone, correlate more closely with mortality than BMI category.^{42–45} Their uptake has been limited by the time required to complete them in a standard consultation.^{46, 47}

3 Epidemiology

Age-standardised prevalence among adults has approximately doubled since 1990.^{13, 48–51} Modelling projects that the number of adults above the class I threshold will approach 1.9 billion by 2035 under current trajectories, with

the steepest proportional increases in low- and middle-income settings rather than in the high-income countries where prevalence is currently highest.^{52–56}

The urban–rural gradient has reversed in several high-income countries, where rural prevalence now exceeds urban.^{57–59} Within-country variation by income decile frequently exceeds between-country variation, a pattern obscured by national reporting.^{48,57,60–63} Childhood trajectories are strongly predictive of adult status, and the tracking coefficient strengthens with age at measurement.^{64–67}

4 Pathophysiology

4.1 Energy homeostasis

Sustained positive energy balance drives triglyceride storage in subcutaneous depots.^{17,68,69} As adiposity increases, afferent leptin and insulin signalling to the arcuate nucleus attenuates, and the defended body-weight set point rises.^{18,70–73} The rise is not readily reversed by energy restriction alone: weight loss provokes a coordinated reduction in resting energy expenditure disproportionate to the loss of lean mass, alongside increased orexigenic drive.^{19,74–77}

This counter-regulatory response persists for at least twelve months after weight stabilisation and may not fully resolve.^{20,78,79} It provides the most parsimonious explanation for the regain trajectories observed across behavioural programmes.^{80–83}

4.2 Adipose tissue expansion and ectopic deposition

Subcutaneous storage capacity is finite and varies substantially between individuals.^{84–86} Once exceeded, lipid deposits ectopically in liver, skeletal muscle, pancreas, and pericardial depots.^{21,87–89} Adipocyte hypertrophy is accompanied by macrophage infiltration and a shift toward a pro-inflammatory cytokine profile, establishing local insulin resistance that propagates systemically.^{90–93}

The variability in storage capacity accounts for the metabolically healthy obese phenotype and its converse, the metabolically unhealthy individual of normal BMI.^{35,94–96} Both argue against anthropometry as a sole risk classifier.

5 Comorbidity Burden

The association with incident type 2 diabetes is the strongest and most consistently replicated, with adjusted hazard ratios above four at BMI exceeding 35.^{97–100} Obstructive sleep apnoea demonstrates a clear dose–response relationship across adiposity strata.^{101,102} Associations with ischaemic heart disease and heart failure with preserved ejection fraction attenuate after adjustment for blood pressure and glycaemia but do not disappear.^{97,103–106}

Musculoskeletal burden is frequently underweighted in risk calculators despite its contribution to disability-adjusted life years.^{107–109} The relationship with depressive

disorders is bidirectional, and the direction of effect differs by age at onset.^{110–112} Weight stigma independently predicts healthcare avoidance and may mediate part of the observed excess mortality.^{113–115}

6 Management

6.1 Lifestyle intervention

Structured programmes produce nadir weight loss of seven to nine per cent at approximately month nine, with partial regain thereafter.^{80,116,117} Approximately sixty per cent of maximal loss is retained at twenty-four months among completers, but attrition reaching thirty-four per cent by month eighteen is the principal limitation of the evidence base.^{81,118,119} Glycaemic benefit in participants with prediabetes persists beyond the weight trajectory.^{98,120}

6.2 Pharmacotherapy

Incretin-pathway agents now achieve mean reductions of twelve to eighteen per cent at sixty-eight weeks.^{22,23,121–123} Gastrointestinal adverse events are the most common reason for discontinuation, and real-world persistence at twelve months falls well below that observed in trials.^{24,124–126} Amylin analogues and combination approaches are at earlier stages of evaluation.^{25,127,128}

Central appetite modulators and absorption inhibitors remain in use where incretin agents are contraindicated, unavailable, or unaffordable.^{26,129} Comparative effectiveness data across classes remain sparse, and most comparisons are indirect.^{130,131}

6.3 Surgical and device-based therapy

Bariatric surgery continues to produce the largest and most durable reductions, with diabetes remission rates exceeding those of any pharmacological comparator.^{105,132,133} Endoscopic approaches occupy an intermediate position in both efficacy and risk.^{134,135}

7 Health Economics and Implementation

Cost attribution methodology varies materially between published estimates. Prevalence-based and incidence-based approaches produce figures differing by a factor of two or more for the same population, and cross-study comparison is inadvisable without harmonisation.^{60,61} Indirect productivity losses are consistently estimated at roughly half the magnitude of direct medical costs, though the confidence intervals are wide.^{62,63}

Implementation remains the binding constraint. Where effective pharmacotherapy exists, access is rationed by cost, by specialist capacity, or by both.^{46,47,113} Care pathways that record the rationale for each escalation step support later audit and appear to improve continuity, though the evidence is observational.^{44,45,114}

8 Conclusions

Obesity is a chronic disease with a well-characterised physiological basis for relapse, a substantial and broad comorbidity burden, and, for the first time, pharmacological options whose effect sizes approach those of surgery. The principal challenge is no longer whether effective treatment exists but whether health systems can deliver it at the duration and scale a chronic disease requires.

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Conflicts of Interest

The authors declare no competing interests. This document is fictional test material produced for software validation purposes.

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